MECH 151

# Technical Design

## Unit Reference Guide

*A student quick-reference for every unit — Design Fundamentals through Design for Manufacturing — for portfolio work, quizzes, and the final prototype.*

Foundations | Unit 1 | Unit 2 | Unit 3 | Unit 4 | Final Exam

COURSE MAP

# Semester Roadmap, Weeks 4–17

*One project-based portfolio module per week; a short foundations unit precedes Unit 1.*

Foundations · Weeks 4–7 · Quiz Wk 7 · Weeks 7–10 · Quiz Wk 10 · Weeks 11–13 · Quiz Wk 13 · Weeks 14–16 · Quiz Wk 16 · Week 17

- **Design Fundamentals**
- **Unit 1 — CAD Fundamentals**
- **Unit 2 — CAD Advanced**
- **Unit 3 — Design for Assembly**
- **Unit 4 — Design for Manufacturing**
- **Final Exam — Prototype Project**

**How grading lines up:** each unit maps to a graded portfolio module and closes with a unit quiz; the Design Portfolio (50% of the course grade) and the Prototype + Final Exam (30%) are graded separately — see the syllabus for the remaining weighting.

COURSE NOTE

# This Course Is CAD-Program-Agnostic

*The field of CAD software is too broad to teach every package — so this course teaches the operations that transfer between all of them.*

**Dozens of parametric CAD packages —** are used across industry — SolidWorks, Fusion 360, Onshape, Inventor, Creo, NX, CATIA, and more

**It's impractical to teach all of them in one course —** and mastering only one would leave you stuck the moment a job, internship, or team uses something else

**So this course teaches the operations every one of them shares —** the specific menu names are yours to look up in whatever software you're using

| The Operation | Sometimes Called |
|---|---|
| Push/pull a 2D profile into a solid | *Extrude · Pad · Boss / Cut* |
| Spin a profile around an axis | *Revolve · Rotate* |
| Lock a sketch so it can't drift | *Relations · Constraints* |
| The running history of modeling steps | *Feature Tree · Timeline · Model Tree* |
| Position parts relative to each other | *Mates · Joints · Assembly Constraints* |
| Round over or angle-cut a sharp edge | *Fillet / Chamfer · Round / Bevel* |

**Why it matters:** everything in this deck — sketching, feature trees, mates, GD&T — describes the operation, not a specific menu path. Whatever software you're using in class, an internship, or a job, look for the tool that does the same job.

FOUNDATIONS

# Design Fundamentals

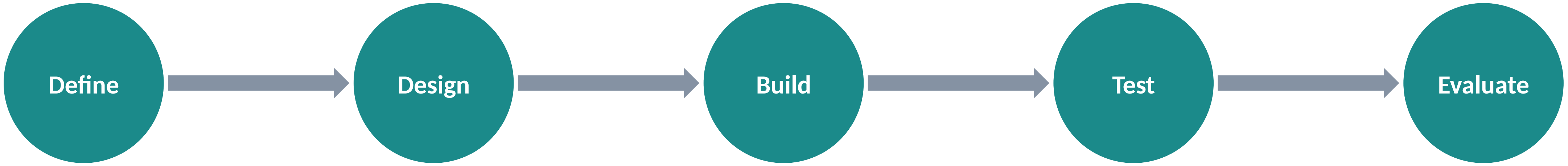
*Step back from the software: what a CAD model is for, before Unit 1's sketches and features begin.*

BEFORE UNIT 1

FOUNDATIONS

# The Design Process Is a Cycle

*Every design loops — the first prototype is expected to be wrong, and that's the process working.*




↺ *then back to Design (or Define), carrying what you learned*

**Define —** state what the part must do, fit, withstand, and cost — before any geometry exists

**Design —** concept options, then the parametric models and drawings that pin down every dimension

**Build —** make a prototype — a print, a machined part, a mock-up

**Test —** measure it, fit it, load it, use it the way it will really be used

**Evaluate —** compare what happened against the requirements; loop back with what you learned

**Why it matters:** if you assume the first version is final, you build a model that's painful to change. If you assume you'll revise it, you build for iteration — clean sketches, captured intent, a readable feature tree.

FOUNDATIONS

# Define the Problem Before You Design

*The step skipped most often — and the one that wastes the most time when skipped.*

| S | M | A | R | T |
|---|---|---|---|---|
| **Specific** | **Measurable** | **Achievable** | **Relevant** | **Time-bound** |
| says exactly what, for whom | has a number you can check | possible with your time & tools | actually serves the need | due by a defined point |

**SMART in practice:** "hold a phone at a comfortable viewing angle" is a direction. "Hold a 600 g phone at 55–70° without tipping, printable in under 4 hours, done by Week 12" is something you can design to and test against.

- Requirements, objectives, and constraints get lumped together as "the spec" — but they filter and rank designs differently (next: how each one works).

FOUNDATIONS

# Requirements, Objectives, Constraints

*Requirements and constraints filter designs to the acceptable ones; objectives rank the acceptable ones to find the best.*

## Requirements

*What the design must do — pass/fail*

- "Supports 5 kg without permanent deflection"
- Either met or not; missing one disqualifies a design
- Verified by test or inspection

## Objectives

*What you're optimizing — directions of "better"*

- "As light as possible," "minimize cost"
- Not pass/fail — compared by degree
- Often trade against each other

## Constraints

*Fixed limits on the solution space*

- "Fits a 150 mm cube," "$25 budget," "due Week 12"
- Not about function — about what's allowed
- Violating one disqualifies the design

**Why it matters:** mix them up and you either reject a good design for missing a "nice to have," or ship one that fails a real must-have.

**Evaluating success —** turn objectives into weighted, measurable criteria and score concepts with a decision matrix — screen pass/fail items first, then rank.

FOUNDATIONS

# Knowing When to Stop: The 80/20 Rule

*No design is ever perfect — chasing perfection is its own failure mode.*

**80/20**

~80% of a design's useful performance comes from the first 20% of the effort — chasing the rest costs far more than it returns.

**Move forward once requirements are met —** even if you can see ways it could be nicer

**Perpetual designing is a cost —** every revision delays the prototype, the test, and everything you'd learn from them

**"Good enough" is a decision —** judge against written requirements, not an imagined ideal

**Hold both ideas at once:** expect the first version to be wrong, and resist the urge to perfect it. The cycle moves forward when a design is good enough to learn from — not when it's flawless, and not before it meets its requirements.

# Design Intent

*The reasoning a model should capture — what you expect to stay true when a dimension changes.*

**Sketch relations —** symmetric, equal, concentric, collinear say why geometry is where it is, so it stays right when dimensions change

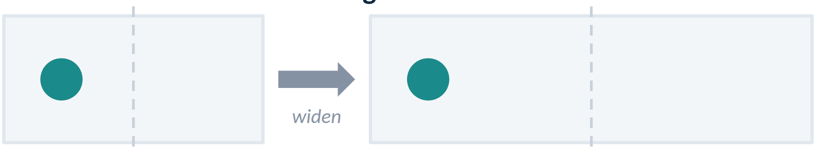
**Which dimensions you choose —** dimension from the reference that actually matters — a mounting face, a mating edge — not whatever's convenient

**Feature order & references —** build later features off stable geometry, not faces that may move or disappear in an edit

**Equations & linked values —** when two dimensions must always match or follow a rule, drive one from the other instead of typing both

*Same result, different intent — dimension a hole two ways:*

**Dimensioned from the left edge:**




**→ widen the block and the hole drifts off-center**

**Tied to center by a symmetric relation:**


**→ widen the block and the hole stays centered**

**Why it matters:** both models look identical until the block changes size — only the second one captures what you actually meant, not just a position that happened to look right.

1 **Editability**

a well-built model updates the way you intend

2 **Communication**

the next person can read the tree and see the logic

3 **Downstream reuse**

assemblies, drawings & manufacturing files inherit it

FOUNDATIONS

# Organizing a Design

*When a design has more than one part, where do the controlling dimensions live?*

## Bottom-Up

*Design each part alone, then assemble*

- Parts are independent — editing one doesn't ripple
- Simple to manage; easy to reuse parts
- Good for well-understood or off-the-shelf parts
- Weak when parts must share dimensions

## Top-Down

*Start from the assembly's shared geometry*

- Shared dimensions live in one place and propagate
- Keeps mating parts fitting as the design evolves
- Good for new designs with moving interfaces
- More setup; unmanaged references get fragile

**Tools for top-down —** layout sketch (fixes key sizes before parts exist), skeleton/master model (shared reference geometry), in-context references (shape one part against another directly).

**Rule of thumb:** use the simplest structure that captures the intent — reach for top-down only where parts genuinely must share dimensions.

FOUNDATIONS

# Practices That Keep a Design Robust

### 1 Fully define every sketch

an under-defined sketch can drift on an unrelated edit

### 2 Reference stable geometry

the origin and default planes never move; a filleted edge might vanish

### 3 Keep sketches simple

one idea per feature — complexity goes in the feature tree

### 4 Name & organize features

a tree of "Extrude7, Cut12" tells the next person nothing

### 5 Minimize dependency chains

fewer things a feature relies on means fewer ways it breaks

### 6 Model the design, not just the shape

if a rule governs the part, build the rule in

UNIT 1 • WEEKS 4–7

# CAD Fundamentals

*Sketching, views, and the first parametric model — the foundation every later unit depends on.*

QUIZ IN WEEK 7

UNIT 1

# There Is No Single "Right" Model

*The same shape can be built from a different base feature, sketch set, or feature order — with effectively infinite valid combinations.*

**What actually separates good from poor:** two classmates can hand in completely different feature trees and both be correct. What separates a good model from a poor one is whether it captures design intent, rebuilds without errors, and stays easy to edit — not whether it matches a canonical answer.

**Focus on —** making deliberate choices you can explain, not on guessing the "expected" answer.

**This unit builds —** the drawing and modeling foundation every later unit depends on — a controlled sketch, standard views, and a parametric solid that updates when you change it.

UNIT 1

# The Origin: Where Every Model Begins

*Every coordinate, sketch, and feature ultimately measures from one fixed point in space.*

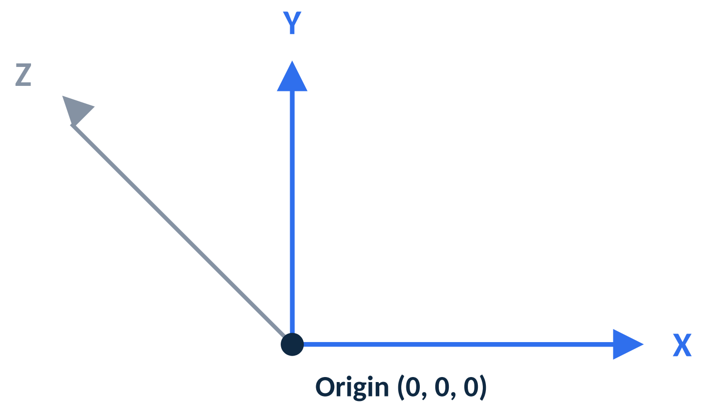
**The origin (0, 0, 0) —** the one point in the file that never moves, no matter how the model is edited

**Three default planes —** Front, Top, and Right all pass through the origin and split space along the X, Y, and Z axes

**Everything is measured from it —** directly or indirectly — a sketch plane, a mate, even a drawing's projected views all trace back to this fixed reference

**Why it matters —** reference the origin (or geometry built from it) instead of an edge that might move, get filleted away, or get deleted in a later edit

*The three default planes meet here*




**Why it matters:** the exact pairing of planes to axes (which plane is "Front") varies by CAD program — but every parametric system anchors its geometry to an origin and a set of default planes the same way.

UNIT 1

# From Point to Solid: Building Up in Dimensions

*Every model is a stack of dimensions — each one built from the last.*

0D

### Point

a location in space — no length, area, or volume. A sketch point, a vertex, the origin itself.

1D

### Line / Curve

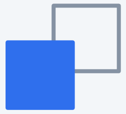
connects two points — has length only. Lines, arcs, and splines: what you actually sketch.

2D

### Polygon (Profile)

a closed loop of 1D entities that encloses an area. A triangle, rectangle, circle — any closed profile.

3D



### Solid (Volume)

push, spin, or sweep a profile through space and it gains volume. Extrude, revolve, sweep, loft.

**Coming up:** the next several slides walk through sketching (0D → 1D → 2D) and solid modeling (2D → 3D) in detail — this is the map for where each one fits.

UNIT 1

# Creating a Body: The Feature Loop

*Every solid feature repeats the same four steps — the feature tree is just this loop, run again and again.*


↺ *repeat, starting the next loop from a face this feature just created*

1 · **Pick a plane —** one of the three default planes, a flat face of the solid so far, or an offset/angled reference plane

2 · **Define a profile —** sketch a closed, fully-defined 2D profile on that plane — points and lines built into a polygon

3 · **Create a volume —** extrude, revolve, sweep, or loft the profile into a solid, then add, cut, or intersect it with what's already there

4 · **Dress it up —** apply fillets, chamfers, shells, drafts, or patterns — operations on the solid, not a new sketched volume

**Why it matters:** step 1 of the next loop is usually a face the previous loop just created, not a default plane — that dependency is exactly what builds the feature tree, one branch at a time.

UNIT 1

# Sketching Fundamentals

*A sketch is a 2D profile on a plane or face — the input to a 3D feature like an extrude or revolve.*

### 1 Sketch entities

lines, arcs, circles, rectangles, slots, splines, and points

### 2 Reference geometry

the origin, the three default planes (front, top, right), and construction lines that guide without becoming part of the profile

### 3 Closed vs. open profiles

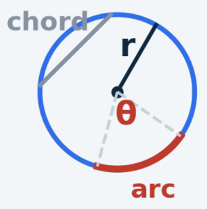
a closed loop can be extruded into a solid; an open profile makes thin features, surfaces, or paths

**Why it matters:** keep sketches simple — draw only the geometry the feature needs, keep profiles closed, and put complexity into separate features rather than one crowded sketch.

UNIT 1

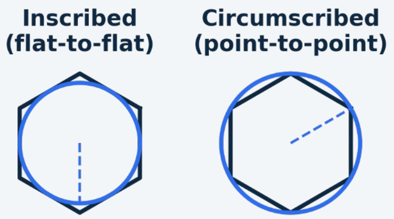
# Geometry of Shapes: 2D Construction

*Before a shape is a sketch, it's plane geometry — points, lines, and curves in defined relationships.*



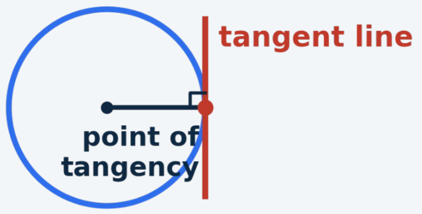

### Circle & Arc Geometry

center + radius/diameter; a chord connects two points, a central angle subtends an arc.



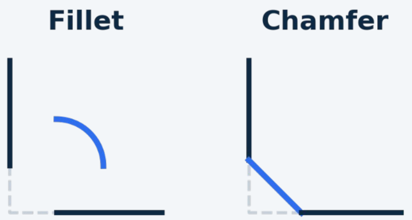

### Polygons

inscribed (flat-to-flat) vs. circumscribed (point-to-point) — matters for hex features and bolt heads.



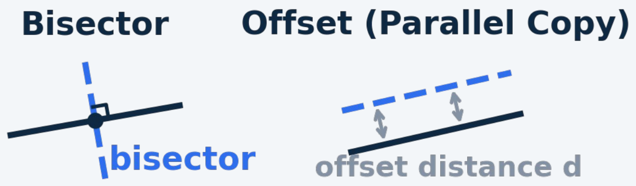

### Tangency

a line/arc touches a circle at exactly one point, perpendicular to the radius there.



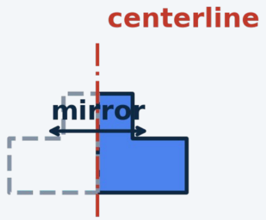

### Fillets & Chamfers

a fillet is a tangent arc replacing a sharp corner; a chamfer is a straight cut across it.




### Constructions

perpendicular bisector and offset — classic constructions, now done by relations.




### Symmetry & Centerlines

build half a shape about a construction line and mirror it to guarantee identical halves.

UNIT 1

# Sketch Constraints & Relations

*Constraints lock down where sketch geometry can move, removing degrees of freedom until the sketch is fully defined.*

### Under-defined

- Geometry can still be dragged
- The sketch is not fully controlled

### Fully defined

- Every entity fixed by relations + dimensions
- The target state before finishing a sketch

### Over-defined

- Conflicting or redundant constraints
- Must be resolved, not ignored

**Geometric relations —** coincident, collinear, parallel, perpendicular, tangent, concentric, equal, horizontal, vertical, symmetric

**Dimensions —** linear, radial, diameter, angular

**Why it matters:** add relations before dimensions — relations capture intent (this hole stays centered) so changing one dimension updates the model predictably instead of distorting it.

UNIT 1

# Orthographic Projection & Multiview Theory

*Imagine the part inside a glass box — each face of the box catches one view; unfolding it gives the standard view arrangement.*

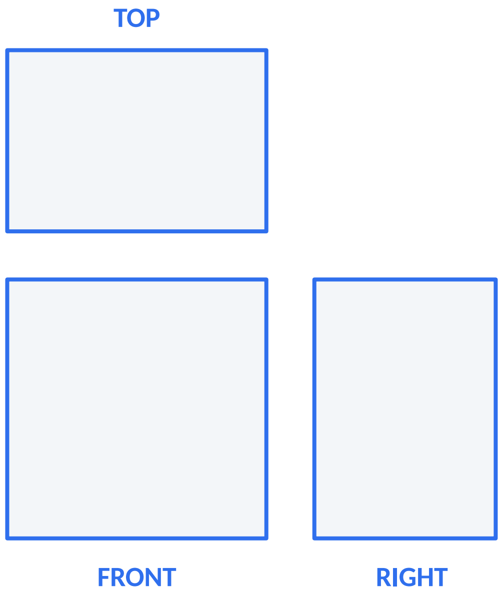
**Third-angle projection —** the U.S. standard — top view sits above front, right-side view sits to its right

**Line conventions —** visible edges solid, hidden edges dashed, centerlines for holes and symmetry axes

**View selection —** choose the front view with the most shape and fewest hidden lines; add only the views needed

**Descriptive geometry —** true length appears only when a line is parallel to the viewing plane; true shape only in a view perpendicular to that plane

**Primary vs. secondary views —** a primary view projects from a principal view; a secondary view projects from a primary view to reveal a feature still not true-size



UNIT 1

# Basic Dimensioning Standards

*Dimensions tell the reader how big the part is and how precisely each feature must be made (ASME Y14.5 in the U.S.).*

### 1 Anatomy

extension lines, dimension lines, arrowheads, and the value — kept off the outline and out of the view

### 2 Placement

dimension to visible outlines, not hidden lines; group related dimensions; smallest closest to the view

### 3 Don't over-dimension

each feature is located once — a redundant dimension creates conflict on the shop floor

### 4 Feature callouts

diameter, radius, counterbore, countersink, and depth symbols for holes

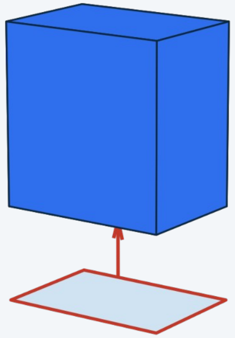
UNIT 1

# Geometry of Shapes: Building 3D Solids

*Every solid model is a running total: start with a volume, add more, and cut some away.*

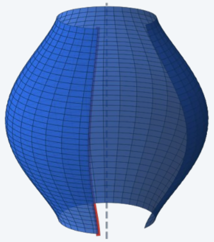
## Extrude

push a profile along an axis



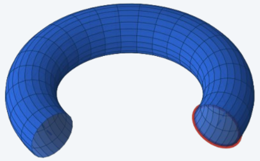
## Revolve

spin a profile around an axis



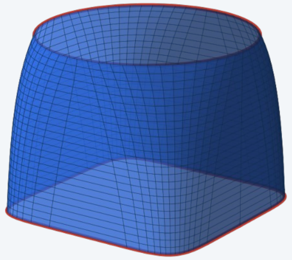
## Sweep

run a profile along a path

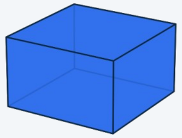


## Loft

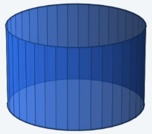
blend between different profiles



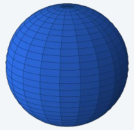
## Primitives



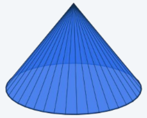
**Block**



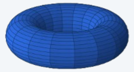
**Cylinder**



**Sphere**



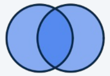
**Cone**



**Torus**

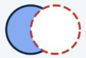
## Booleans & Dress-Up

**Union**



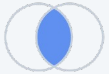
combine A + B

**Subtract**



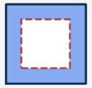
remove B from A

**Intersect**



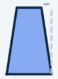
keep only overlap

**Shell**



hollow, keep wall t

**Draft**



taper wall by angle

**Pattern**

repeat at spacing d

*Fillet & chamfer are shown on the sketching slide (2D construction).*

UNIT 1

# Introduction to Parametric Part Modeling

*A parametric model is a history of features, each driven by dimensions and relations — change a dimension and the model rebuilds.*

**Base feature —** the first solid, usually an extrude or revolve of a fully defined sketch

**Added features —** extrude, cut, revolve, fillet, chamfer, hole, shell, pattern — each added in order on the feature tree

**Sketch planes & references —** choosing where each feature's sketch lives so edits stay stable

**Practice —** editing driving dimensions, reordering/suppressing features, fixing rebuild errors — reading a feature tree and predicting what an edit will do

**Why it matters:** this is what makes CAD editable rather than something you redraw from scratch.

UNIT 1

# Geometry of Shapes: Measurable Properties

*Every shape carries numbers that follow from its geometry — the values a design is checked against.*

## Of a 2D profile

- Area, perimeter, centroid
- Second moment of area (I) — resistance to bending
- Section modulus (I ÷ distance to extreme fiber)

## Of a 3D solid

- Volume, surface area
- Mass (volume × density), center of gravity
- Mass moment of inertia — resistance to angular acceleration

**Garbage in, garbage out:** the numbers are only right if the material density is set correctly and the model is a single watertight solid — an open or multi-body model gives a meaningless mass.

UNIT 1

# Unit 1 Wrap-Up

1. **Controlled sketches**

   fully defined, with relations that capture intent

2. **Multiview drawing**

   correct projection, line types, and dimensions

3. **First parametric model**

   a clean, editable feature tree

✓ **You'll produce:** controlled sketches, a multiview drawing, and a first fully parametric part model.
**Assessed by:** Unit 1 Quiz (Week 7) and the corresponding portfolio modules.

UNIT 2 • WEEKS 7–10

# CAD Advanced

*Detailed views, fasteners, and production drawings — turning a model into a document a shop can build from.*

QUIZ IN WEEK 10

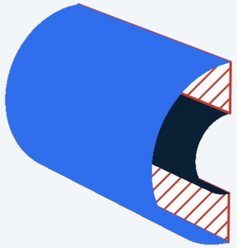
UNIT 2

# Section Views

*An imaginary cutting plane slices the part to show what's inside, replacing hidden-line tangles with a clear cut face.*

## Full Section

The cutting plane passes entirely through the part, exposing the interior along its whole length.



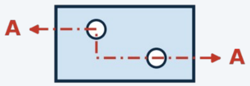
## Half Section

Only one quarter is removed — the exterior and the interior appear together in a single view. Best for symmetric parts.

## Offset Section

The plane is stepped (jogged) so it passes through features that aren't in line — each still appears true size.

**Top view — stepped cutting plane**

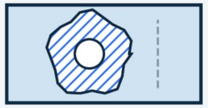



**Section A-A — both holes shown true size**

## Local Sections

Broken-out, revolved, and removed — used to show one local feature without a full cut.

**Broken-out**

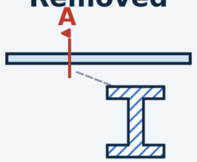


torn boundary reveals only the local interior

**Revolved**

true cross-section rotated 90° into the view, in place

**Removed**




SECTION A-A

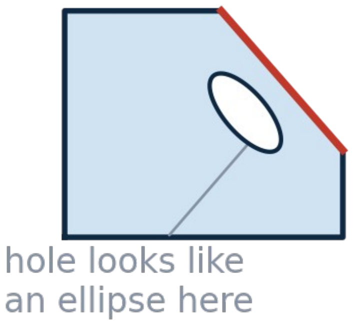
**Convention:** section lining marks the cut face — but ribs, webs, fasteners, shafts, and pins are never hatched, even when the cutting plane passes through them.

UNIT 2

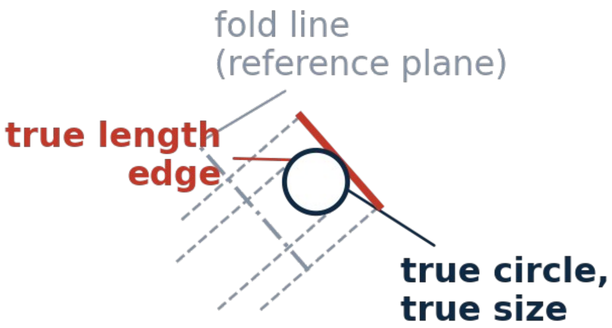
# Auxiliary Views

*An inclined face looks foreshortened in every principal view — an auxiliary view is projected perpendicular to it so it appears true-size.*

**Front view — inclined face foreshortened**




**Primary auxiliary — projected true size**

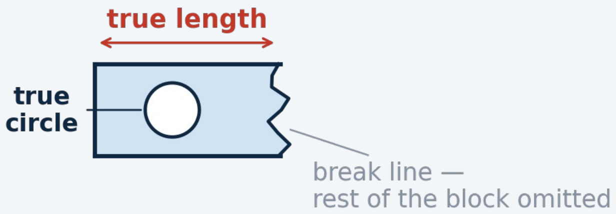



**Primary auxiliary** — projected from one principal view to show an inclined surface true-shape

**Partial auxiliary** — only the inclined feature is drawn — the rest is omitted to avoid distortion and clutter

**Reference-plane method** — measurements are transferred across the fold line along projection lines

**Partial auxiliary — only the true-shape feature is drawn**



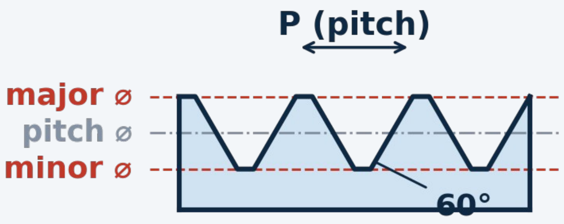
UNIT 2

# Threads & Fastener Representation

*Threads are almost never drawn in full detail — a standardized representation plus a note carries the information a shop needs.*

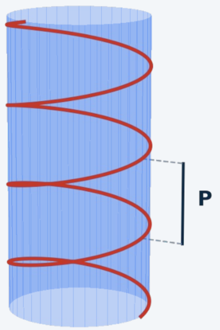
## Thread Profile

**Thread profile (external V-thread)**




## How a Thread Is Cut

**One helical path, advancing one pitch per turn**




**Thread terms** — major/minor diameter, pitch, lead, external vs. internal, right- vs. left-hand, class of fit

**Representation** — detailed, schematic, simplified — simplified is most common on production drawings

**Fasteners** — bolts, screws, nuts, washers, inserts — clearance and tap-drill holes are sized to fit them

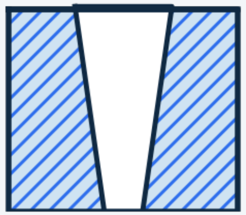
**Thread notes** — 1/4-20 UNC-2A (inch) or M6 × 1 (metric) — diameter, pitch, and class in one callout

## Common Hole Types

**Through Hole**

passes all the way through

**Tapered Hole**



diameter changes along its length

**Blind Hole**

stops before the far side (drill point)

**Counterbore**

flat-bottomed step for a bolt head

**Countersink**

conical seat for a flat-head screw

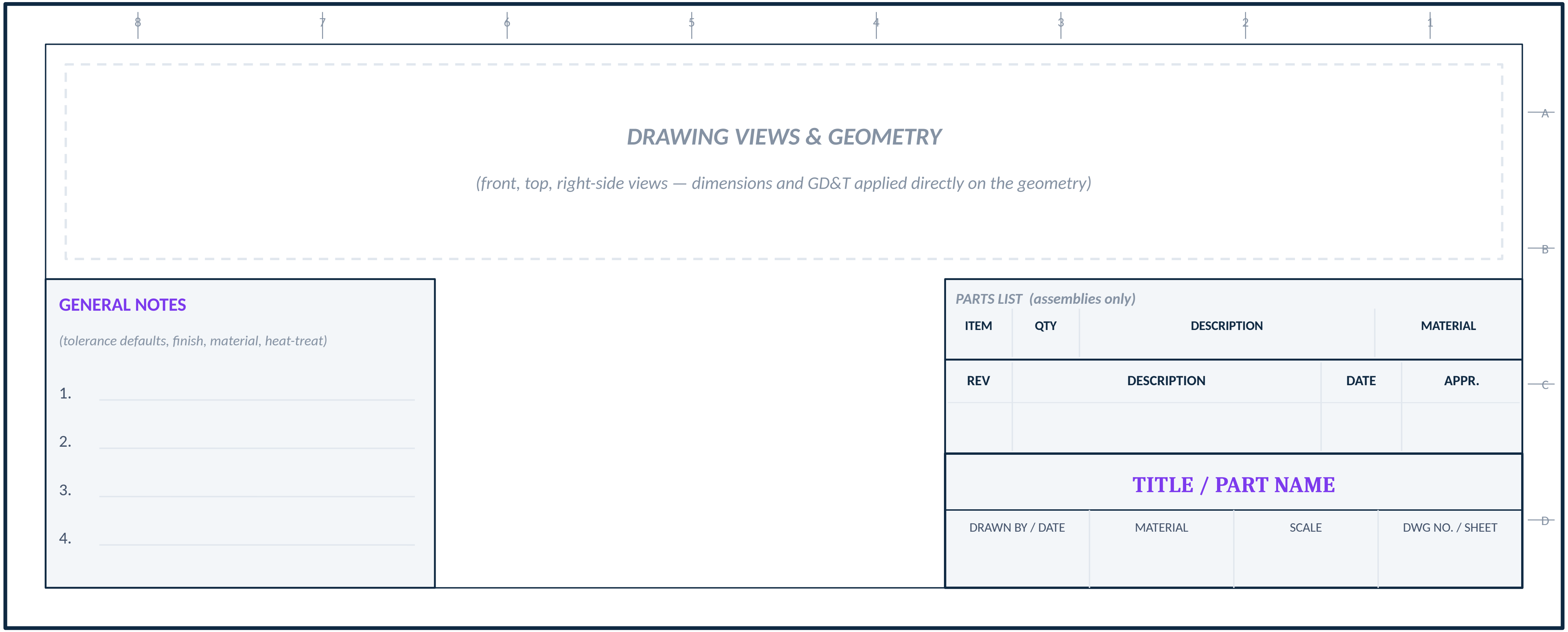
**Spotface**

shallow flat land for a bearing surface

UNIT 2

# Production & Detail Drawings

*A detail drawing is a controlled document — its format carries information that isn't in the geometry.*




*A blank sheet layout — every real drawing fills these same zones with the part's views, dimensions, and notes.*

UNIT 2

# Introduction to Tolerancing

*No part is made to an exact dimension — tolerancing states how much variation is acceptable so parts still fit and function.*

| | | |
|---|---|---|
| ⌀ | **Diameter** | *⌀25* |
| R | **Radius** | *R5* |
| ± | **Plus/Minus** | *±0.01* |
| SQ | **Square** | *SQ20* |
| SR | **Spherical Radius** | *SR10* |
| ° | **Degrees** | *45°* |
| $\boxed{25}$ | **Basic Dimension** | *[25]* |
| ( ) | **Reference Dim.** | *(25)* |

### Limit Dimensions

$$\frac{1.253}{1.247}$$

*high limit over low limit — no math on the shop floor*

### Plus/Minus Tolerance

1.250 ±0.003

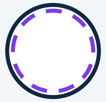
*bilateral shown; unilateral: +0.005 / −0.000*

### GD&T Callout

| ⌖ | ⌀0.10 Ⓜ | A |
|---|---|---|

*position, ⌀0.10 at MMC, to datum A*

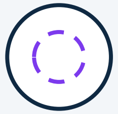
### Close Fit



⌀25 H7/p6

*near-zero gap — press or light-tap assembly (interference)*

### Loose Fit



⌀25 H11/c11

*visible gap — free-running clearance, easy to assemble*

### Surface Finish


*roughness average (μin/μm) — smaller number = smoother surface*

UNIT 2

# CAD Software Configuration

*Consistent output comes from configured software, not from fixing every file by hand.*

### 1 Templates

part, assembly, drawing templates with the right units, standard, title block

### 2 Document properties

units, dimensioning standard (ASME/ISO), grid, layers

### 3 Custom properties

file metadata that auto-populates the title block and BOM

### 4 File management

naming, folder structure, keeping references intact

UNIT 2

# CAD/CAM Data Translation

*Moving a model between systems means exporting to a neutral format — each keeps some information and loses the rest.*

### Solid exchange

*STEP (AP203/214/242), IGES*

- Geometry preserved
- Parametric feature history is not

### Kernel formats

*Parasolid, ACIS*

- Used between packages sharing a modeling kernel

### Mesh formats

*STL, 3MF*

- Faceted surface, no true curves
- Resolution set at export

**Why it matters:** lost or corrupted faces, gaps requiring geometry healing, and unit-scale errors are the common translation issues.

UNIT 2

# Unit 2 Wrap-Up

*Freedom in modeling, discipline in documentation — how you build the part stays open-ended; the drawing follows shared conventions (ASME Y14.5, ISO) so anyone can read it.*

### 1 Documentation-ready drawing

section & auxiliary views, complete title block

### 2 Thread & fastener callouts

with dimensional tolerances applied

### 3 Configured templates

and a clean, translated model export

✓ **You'll produce:** a fully detailed production drawing with section & auxiliary views, fastener callouts, tolerances, plus a correctly configured template and clean export.

**Assessed by:** Unit 2 Quiz (Week 10) and the corresponding portfolio modules.

UNIT 3 · WEEKS 11–13

# Design for Assembly

*Combine part models into working assemblies, document them, and design them to be quick and cheap to put together.*

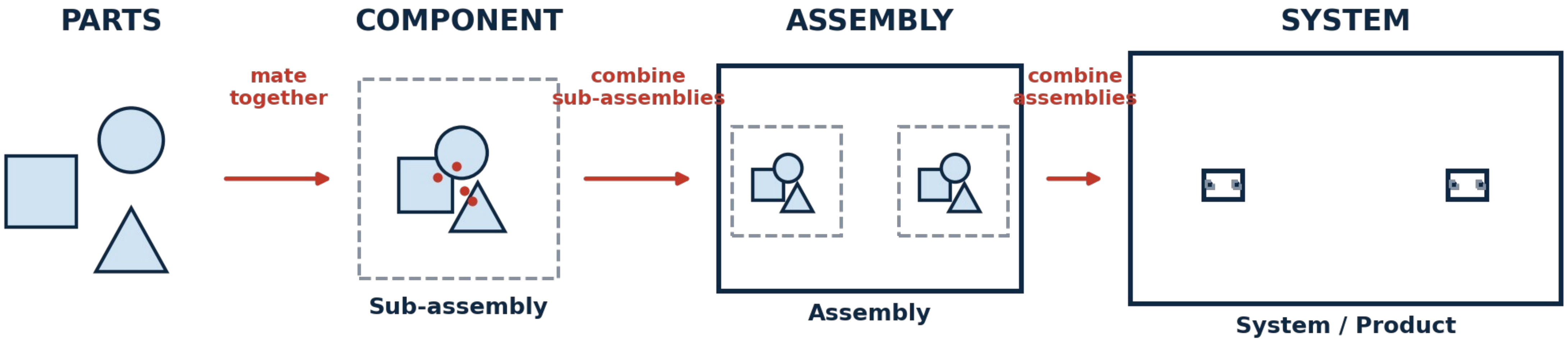
QUIZ IN WEEK 13

03

UNIT 3

# From Parts to Systems

*The same relationship repeats at every scale: individual parts mate into a component, components combine into an assembly, and assemblies combine into a complete system.*




**Why break it down:** this is how large, complicated systems stay manageable — a vehicle or an instrument is never modeled as one flat list of thousands of parts. It's a tree of assemblies, each one a self-contained, testable unit.

UNIT 3

# Assembly Modeling: Mates & Constraints

*An assembly positions components relative to one another using mates — each removes degrees of freedom.*

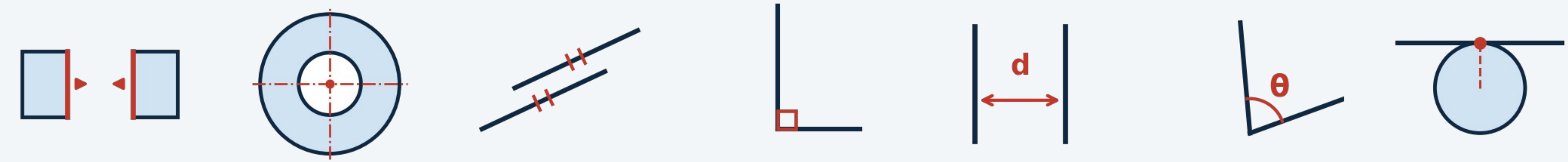
**6**

degrees of freedom a free rigid body has — mates remove them one axis at a time.

**First component —** the first part placed is fixed and becomes the ground the rest is built against

**Motion —** leaving rotational or linear freedom on purpose so a mechanism can be moved and checked

| Coincident | Concentric | Parallel | Perpendicular | Distance | Angle | Tangent |
|---|---|---|---|---|---|---|
| faces (or edges) forced flush | shared axis — pin in a hole | edges stay equidistant | edges held at 90° | fixed gap between faces | fixed angle between edges | curve just touches a line |



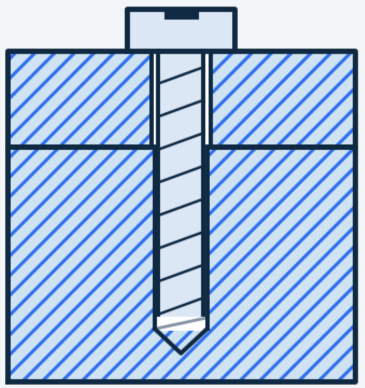
UNIT 3

# Methods of Joining, Part 1

*How two parts are held together is a design decision — choose deliberately, then design both parts to suit it.*

## Threaded Fasteners

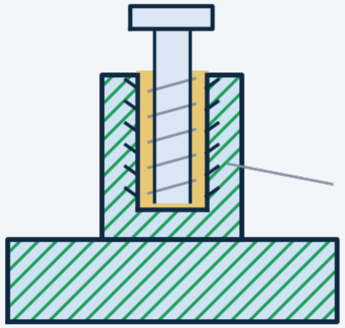
*Into tapped holes*




Default for metal-to-metal joints — strong, standard, and removable. Needs ~1.5–2× screw diameter of thread engagement plus tool access.

## Heat-Set Inserts

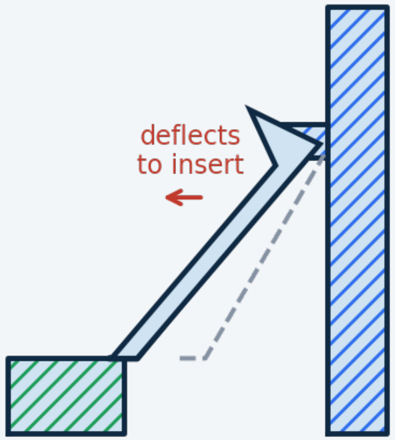
*For plastic / 3D-printed parts*




A brass insert is pressed into a boss with a soldering iron — melted plastic locks into its knurls, far more reusable than threading into plastic directly.

## Snap-Fit Joints

*No hardware, no tools*



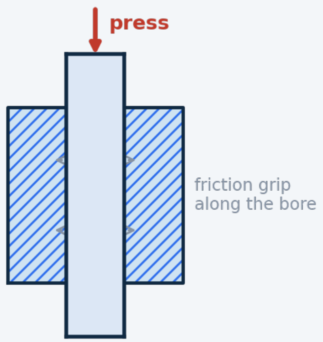

Cantilever, annular, and torsion snaps release or lock permanently depending on retention angle — design for allowable strain and a lead-in angle.

UNIT 3

# Methods of Joining, Part 2

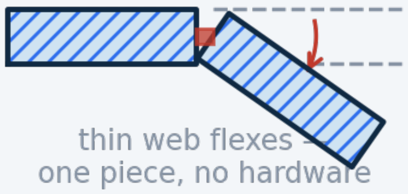
## Press / Interference Fits

Hold by friction from a slight size mismatch between mating diameters — dowel pins, bushings, bearings. Cheap and precise, but hard to disassemble and tolerance-sensitive.



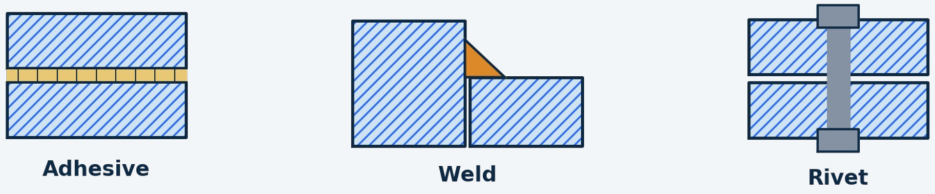

## Compliant Mechanisms

Motion or retention comes from the material's own flexibility — living hinges, flexure pivots, one-piece clips. Cuts part count to the extreme; limited travel and fatigue life.




## Permanent Joints

*Adhesives, solvent/ultrasonic welding, and riveting — used only where the joint never needs to separate.*




**The DFA trade-off:** hardware is universal and serviceable but adds parts, tools, and assembly time; snap-fits and compliant features add almost nothing to the parts list but must be designed carefully and are harder to revise once tooling is dialed in.

UNIT 3

# Using Online Component Repositories

*You should not model a part you can download — this is normal professional practice, not a shortcut.*

**Where models come from —** McMaster-Carr (download STEP/native from almost any product page), manufacturer sites, TraceParts, GrabCAD, 3D ContentCentral

**Why it matters for DFA —** a downloaded model is the real catalog part — correct dimensions and thread, a real part number for the BOM, at the price and lead time listed

**Bring them in cleanly —** keep in a dedicated library folder, don't edit their geometry, mate to them like any other component

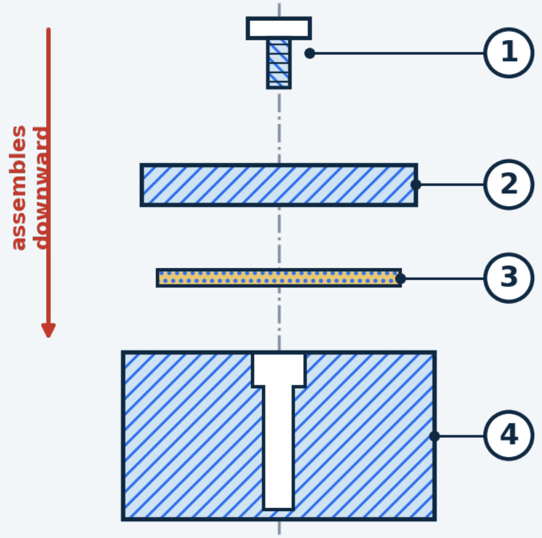
**Check before you trust —** confirm units/scale on import, simplify heavy models, verify the part number matches what you intend to buy

UNIT 3

# Assembly Drawings

*An assembly drawing shows how parts go together — exploded, numbered, and tied to a parts list.*

**Exploded view — balloons keyed to the BOM**




*Balloon numbers are assigned in assembly order and match the item numbers in the parts list at right.*

## Bill of Materials

| ITEM | PART NO. | DESCRIPTION | QTY |
|---|---|---|---|
| 1 | HW-0425 | Socket head cap screw, M4 × 12 | 4 |
| 2 | 100-002 | Cover plate | 1 |
| 3 | 100-003 | Gasket, silicone | 1 |
| 4 | 100-001 | Housing (base) | 1 |

In CAD, this table is generated directly from the assembly and stays linked — edit the model and the parts list updates automatically.

UNIT 3

# Design for Assembly (DFA) Principles

*Fewer parts, easier insertion, less to get wrong on the line — DFA turns assembly time into a design variable.*

### 1 Minimize part count

a part is a candidate for elimination unless it moves, needs a different material, or must be removable

### 2 Prefer off-the-shelf parts

cheaper, faster, and more reliable than custom-fabricated — see the online repositories slide

### 3 Easy insertion & no reorientation

chamfers and self-locating features, assembling from one direction

### 4 Standardize fasteners & use symmetry

fewer unique sizes to stock and pick; make orientation obvious or irrelevant

**Design review & iteration:** the unit ends with a design review workshop — present an assembly, take feedback against the DFA principles, and produce a documented revision with a measurable improvement.

UNIT 3

# Unit 3 Wrap-Up

### 1 Multi-part assembly

correct mates, sub-assemblies, and library hardware

### 2 Exploded drawing + BOM

balloons keyed to a linked bill of materials

### 3 Documented revision

a design review turned into a measurable DFA improvement

✓ **You'll produce:** a multi-part assembly model with correct mates, an exploded drawing with balloons and a linked BOM, plus a documented design revision.
**Assessed by:** Unit 3 Quiz (Week 13) and the corresponding portfolio modules.

UNIT 4 · WEEKS 14–16

# Design for Manufacturing

*Close the loop from CAD model to physical part — manufacturing-ready files, process limits, and shop safety.*

QUIZ IN WEEK 16

UNIT 4

# Why Standards Carry the Design Intent

*Once a design leaves your screen, the maker builds exactly what the documentation says.*

**The maker is not the designer —** standard conventions (ASME Y14.5, GD&T, thread/finish callouts) let them read your intent without a phone call

**Critical dimensions must be stated —** a tolerance callout tells the shop where to spend effort and where a loose result is fine

**Standard formats preserve meaning —** STEP carries clean geometry between systems; a sloppy export arrives with gaps, wrong units, or lost features

**Ambiguity is expensive and slow —** on a graded prototype, it comes back as a part you can't use and no time to redo

**Why it matters:** capture the design intent, then communicate it in the form everyone downstream already knows how to read.

# GD&T: The Building Blocks

*A symbolic language (ASME Y14.5 / ISO GPS) for functional relationships — design intent written in symbols instead of prose.*

**Feature of size —** a feature with an axis or centerplane — a hole, pin, slot, width

**Datums (A, B, C) —** reference surfaces chosen to match how the part actually seats in use

**Datum reference frame —** primary + secondary + tertiary datums lock all six degrees of freedom

**Tolerance zone —** the geometric space the feature must lie within — a gap, a cylinder, a band

**Symbol families —** the standard's ~14 geometric characteristic symbols sort into five families — form, orientation, location, profile, and runout (next slide)

*Feature control frame — read left to right*

| ⌖ | ⌀0.2 M | A B C |
| --- | --- | --- |
| Symbol (position) | Tolerance value | Datums referenced |

**Why it beats plain ±:** a ± box on four hole dimensions creates a small square zone and over-constrains the part; a position callout defines a round zone tied to the surfaces that matter, letting more good parts through.

UNIT 4

# GD&T Symbol Categories

*The standard's ~14 geometric characteristic symbols sort into five families by what they control and whether they need a datum.*

## Form

*No datum needed*

- Straightness
- Flatness
- Circularity
- Cylindricity

## Orientation

*Needs a datum*

- Perpendicularity
- Parallelism
- Angularity

## Location

*Needs datums*

- Position
- Concentricity
- Symmetry

## Profile & Runout

*Profile: no/some datums · Runout: datum axis*

- Profile of a line
- Profile of a surface
- Circular runout
- Total runout

**Material condition modifiers:** MMC (maximum material condition) and LMC (least material condition) let a position tolerance earn bonus tolerance as the hole moves away from its worst-case size — matching how a bolted clearance-hole joint really behaves.

UNIT 4

# Manufacturing-Ready Files

*Each process consumes a different file — a good part starts with exporting the right one at the right settings.*

### Additive

- Watertight mesh: STL or 3MF
- Fine enough to hide facets, not so fine the file is unwieldy

### Laser Cutting

- 2D vector profile (DXF)
- Cut and engrave lines on separate layers

### CNC Machining

- Solid model (STEP) + dimensioned drawing
- Drawing tells the machinist which dimensions are critical

**CAM basics —** stock setup, toolpath generation, simulation, and the post-processor that turns toolpaths into machine G-code.

UNIT 4

# Machine Safety, PPE & Operation

*Students are checked out on safe setup and operation before running any shop equipment.*

**Baseline PPE —** ANSI Z87.1 safety glasses, closed-toe shoes, no loose clothing/jewelry/unrestrained hair

**Rotating machinery —** never wear gloves near a spindle, chuck, or bit — a caught glove pulls the hand in

**Laser cutter —** never run unattended; keep a view of the cut; know the extraction and fire response

**3D printers —** hot nozzle and bed — let them cool, don't reach into an active machine

**Before starting —** know the location of the e-stop, fire extinguisher, and first-aid kit

**PPE is not optional:** per the syllabus, a participant without proper PPE during machine use will be asked to leave the class for that session.

UNIT 4

# Additive Manufacturing (3D Printing)

*The course uses material-extrusion printing (FDM/FFF), building one deposited layer at a time.*

**Materials —** PLA, PETG, ABS — printability, strength, and temperature trade-offs

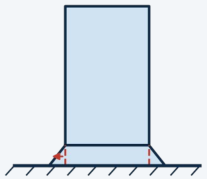
**Orientation —** layers are weakest in the build (Z) direction — a strength decision, not just fit-on-bed

**Supports & overhangs —** the roughly 45° rule, bridging limits, designing to avoid support

**Settings that matter —** layer height, wall count, infill density and pattern

**Tolerances —** holes print undersize, first layers spread — allow for it or drill/ream after

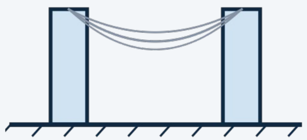
*Common Print Issues & Fixes*



### Elephant Footing

*the first layer over-extrudes and squishes outward under the nozzle*

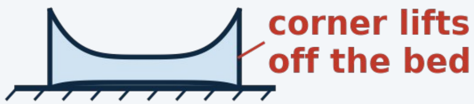
**Fix —** add a brim, or reduce first-layer horizontal expansion



### Stringing

*molten filament oozes from the nozzle while traveling between features*

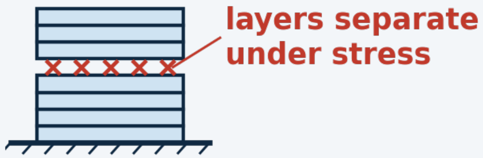
**Fix —** dry the filament; tighten retraction distance/speed




### Warping

*outer edges cool and shrink faster than the center, curling corners up*

**Fix —** add a brim/raft; enclose the printer or use a heated bed




### Layer Separation

*layers don't fully fuse — often too cold, or cooled by a draft*

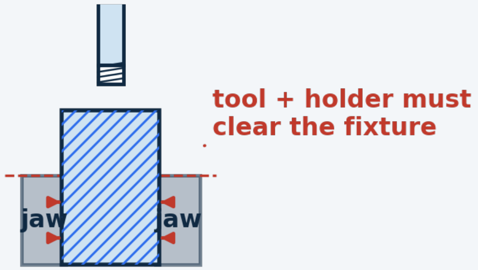
**Fix —** raise nozzle temp slightly; shield the print from drafts

UNIT 4

# Subtractive Manufacturing (CNC)

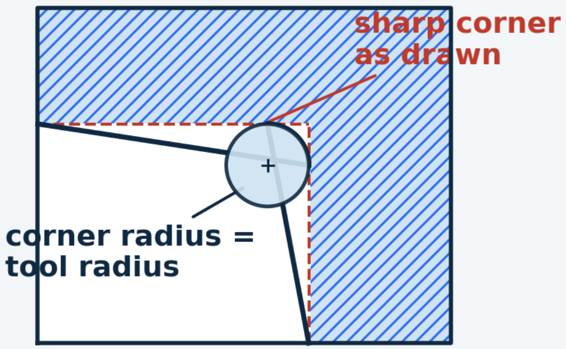
*Starts with solid stock and cuts material away — design constraints come from the tool and how the part is held.*

### Workholding: rigid, and clear of the tool

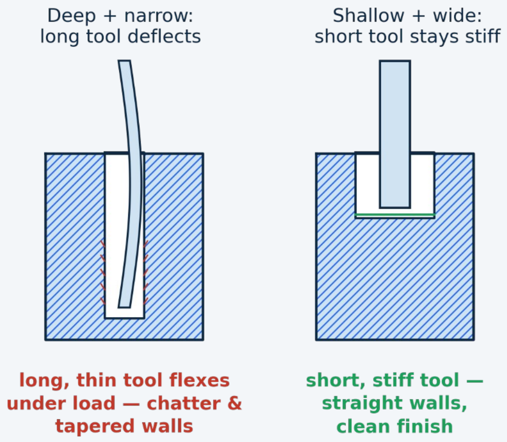



workpiece clamped between vise jaws

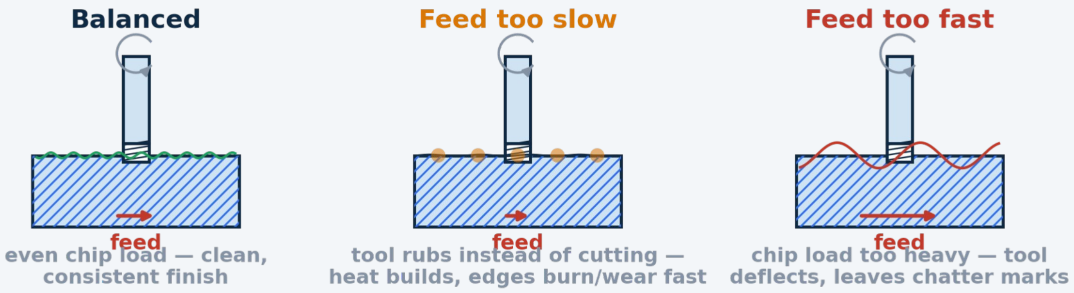
### Internal corners inherit the tool radius




### Tool reach vs. stiffness




*Feeds & Speeds*



UNIT 4

# An Introduction to G-code

*A plain-text list of moves the machine runs one line at a time — the shared language of CNC mills, 3D-printer slicers, and laser cutters.*

- **What it is —** a text file a controller runs top to bottom; G words set motion and modes, M words switch machine functions
- **A line is a block —** a list of words, each an address letter plus a value: X25.0 is address X, value 25.0
- **Modal settings stick —** units, positioning, feed and speed stay in effect until a later line changes them
- **Where it comes from —** CAM or a slicer writes it; a post-processor tailors it to your exact machine — then you set the part zero and dry-run before cutting

*One block, broken down*

```
G01    X25.0  Y10.0    F150
```

| G01 | X25.0 Y10.0 | F150 |
|---|---|---|
| linear feed move (modal) | target coordinates | feed rate, mm/min (modal) |

*Modal words hold: this F150 and G01 keep acting on every line below until one is changed.*

### *Codes you'll actually see*

| | | | | | |
|---|---|---|---|---|---|
| `G20 / G21` | inch / millimetre | `G02 / G03` | clockwise / CCW arc | `M03 / M05` | spindle on CW / off |
| `G90 / G91` | absolute / incremental | `G54–G59` | work offset (part zero) | `M06` | tool change |
| `G17` | work in the XY plane | `G28` | return to machine home | `M08 / M09` | coolant on / off |
| `G00` | rapid — non-cutting move | `G04` | dwell — timed pause | `M30` | end of program |
| `G01` | linear move at feed rate | `M00` | pause for the operator | `S / F / T` | speed / feed / tool no. |

UNIT 4

# Working with G-code

*Small edits at the edges of a program are routine — editing the geometry in the middle is CAM's job.*

*A short program, annotated*

```
(SLOT — 6MM 2-FLUTE ENDMILL)
G21 G90 G17    mm, absolute, XY plane
G54            work offset = part zero
G0 Z5.0        rapid to safe height
M3 S10600      spindle on CW, 10600 rpm
G0 X0 Y0       rapid above the start
G1 Z-2.0 F300  feed down to depth
G1 X50.0 F640  cut the slot at feed
G0 Z5.0        retract
M5             spindle off
M30            end of program
```

## Safe to edit by hand

- Feed (F) and spindle speed (S) — ease off for chatter or burning
- Safe / clearance Z heights — after checking nothing is taller
- Work offset (G54→G55); add a pause (M00) or dwell (G04)
- Trim header/footer lines your machine doesn't use; drop an unused tool section

## Regenerate from CAM — don't hand-edit

- Individual X / Y / Z cutting values, or arc I / J centres
- Switching units (G20 ↔ G21) partway through
- Cutter compensation (G41 / G42) offsets

**Prove it before it cuts:** re-simulate the whole program, then run it once as an air pass with a hand on the feed-hold — a G-code error is found by the machine at full speed.

UNIT 4

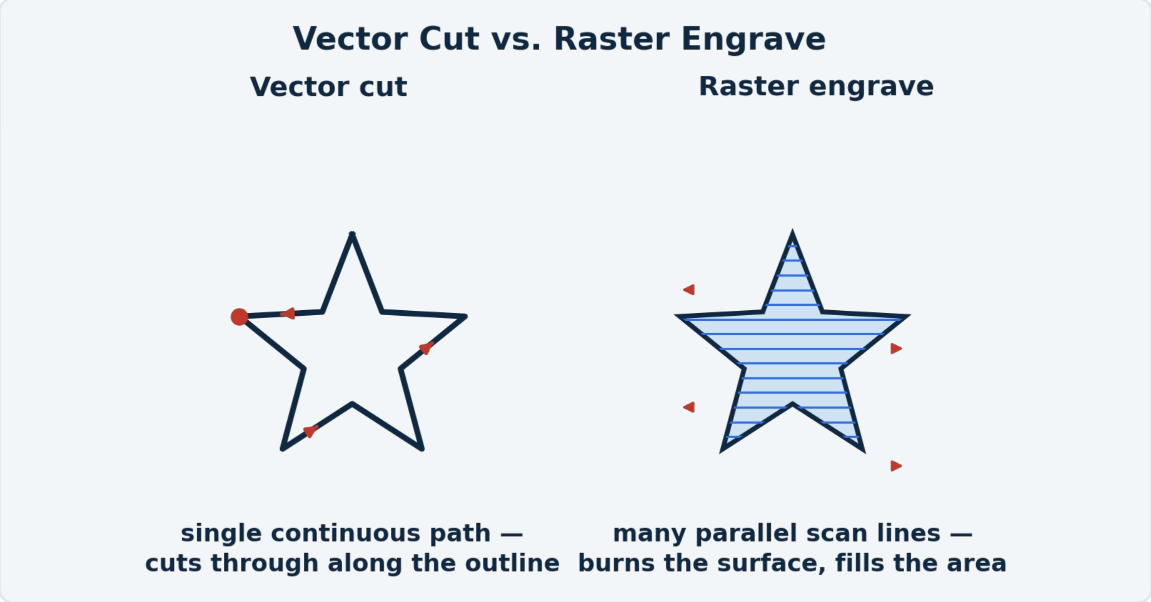
# Laser Cutting

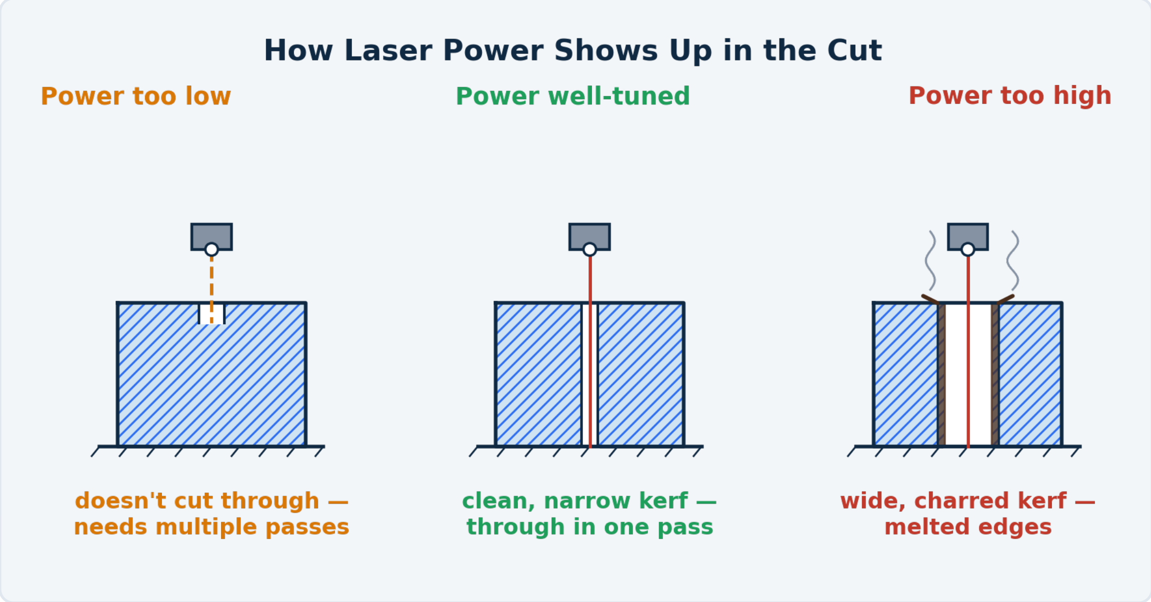
*Cuts flat sheet by burning or vaporizing a narrow line through it — fast and precise in 2D, with its own rules.*

**Kerf —** the cut removes a finite width of material — account for it on tight-fitting parts

**Material limits —** wood, acrylic, paper, and cloth cut well

**Small-feature limits —** minimum hole size and spacing relative to material thickness




**Never cut:** PVC and other chlorinated plastics must never be cut on a laser — they release chlorine gas.

UNIT 4

# DFM Design Considerations

*Shape the part so the chosen process can make it well, at reasonable cost, without special effort.*

### 1 Match design to process

uniform walls for printing/casting, generous radii for machining, flat unfolded geometry for laser

### 2 Minimize setups

features on one face are cheaper than features needing re-fixturing

### 3 Tolerance only what matters

tight tolerances drive cost — apply to mating/functional features

### 4 Use standard stock & tooling

standard sizes and drill/tool radii cost less, arrive faster

### 5 Design for prototype AND product

what's fine for a one-off print may not survive production

UNIT 4

# Unit 4 Wrap-Up

### 1 Manufacturing-ready files

correctly exported for the chosen process

### 2 GD&T documentation

for at least one process, showing what you'd control and why

### 3 A DFM-reviewed design

ready to hand off to a shop process

✓ **You'll produce:** manufacturing-ready files and GD&T documentation demonstrating your design-for-manufacturing reasoning, plus reading and safely editing a G-code file.
**Assessed by:** Unit 4 Quiz (Week 16). The physical prototype build & presentation is graded separately as the course Final Exam.

QUICK REFERENCE

# All Units at a Glance

*Keep this page open while you work — every unit's timing, focus, and assessment in one place.*

| | | | |
|---|---|---|---|
| **Foundations** | Before Unit 1 | Define the problem, design intent, top-down vs. bottom-up | — |
| **Unit 1 — CAD Fundamentals** | Weeks 4–7 | Controlled sketches, multiview drawing, first parametric model | Quiz, Wk 7 |
| **Unit 2 — CAD Advanced** | Weeks 7–10 | Section/auxiliary views, threads, tolerancing, production drawing | Quiz, Wk 10 |
| **Unit 3 — Design for Assembly** | Weeks 11–13 | Mates, joining methods, exploded drawing + BOM, DFA principles | Quiz, Wk 13 |
| **Unit 4 — Design for Manufacturing** | Weeks 14–16 | GD&T, manufacturing files, G-code, additive/subtractive/laser, DFM, safety | Quiz, Wk 16 |
| **Final Exam — Prototype Project** | Week 17 | Team-built physical prototype, manufacturing docs, presentation | 30% (w/ prototype) |